

TCN_MHA (legacy production model)

The previous production model: causal TCN with multi-head attention, judged as one more candidate.

The current production model, evaluated as one more candidate (not as a reference). It is not deployable on the microcontroller and is beaten by tcn_lite in almost every subject.

Architecture

- Multi-scale causal convolution stem (kernels 3, 5 and 9, 16 filters each, GELU), concatenated to 48 channels.
- Four residual blocks of two causal Conv1D(48, kernel 3) with LayerNormalization, GELU and SpatialDropout(0.1), dilations 1, 2, 4 and 8.
- MultiHeadAttention over the 30 time steps (4 heads, key dim 12) with a residual connection and LayerNorm, then GlobalAveragePooling1D.
- Static-feature branch: Dense(32, GELU) + LayerNorm, used to modulate the pooled sequence (multiply and add gates) before a Dense(64, GELU) + LayerNorm, Dense(32, GELU), Dense(1) head.
- 77,345 parameters (about 75.5 KB int8) and about 2,040,576 MACs per prediction; not streamable.
- Trained in the legacy pipeline with RSAPhaseAwareLoss: Huber + a sign penalty + a variance-matching penalty. The variance term rewards oscillation, a plausible cause of its invented RSA peaks.

Layer table (B = batch)

layer	configuration	output shape	params
InputLayer		(B, 30, 1)	0
Conv1D	filters=16, kernel_size=3, activation=gelu, padding=causal	(B, 30, 16)	64
Conv1D	filters=16, kernel_size=5, activation=gelu, padding=causal	(B, 30, 16)	96
Conv1D	filters=16, kernel_size=9, activation=gelu, padding=causal	(B, 30, 16)	160
Concatenate		(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Add		(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, dilation_rate=2, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, dilation_rate=2, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Add		(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, dilation_rate=4, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, dilation_rate=4, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Add		(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, dilation_rate=8, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Conv1D	filters=48, kernel_size=3, dilation_rate=8, padding=causal	(B, 30, 48)	6960
LayerNormalization		(B, 30, 48)	96
Activation	activation=gelu	(B, 30, 48)	0
SpatialDropout1D	rate=0.1	(B, 30, 48)	0
Add		(B, 30, 48)	0
MultiHeadAttention	num_heads=4, key_dim=12	(B, 30, 48)	9408
InputLayer		(B, 12)	0
Add		(B, 30, 48)	0
Dense	units=32, activation=gelu	(B, 32)	416
LayerNormalization		(B, 30, 48)	96
LayerNormalization		(B, 32)	64
GlobalAveragePooling1D		(B, 48)	0
Dense	units=48	(B, 48)	1584
Multiply		(B, 48)	0
Dense	units=48	(B, 48)	1584
Add		(B, 48)	0
Concatenate		(B, 80)	0
Dense	units=64, activation=gelu	(B, 64)	5184

layer	configuration	output shape	params
LayerNormalization		(B, 64)	128
Dense	units=32, activation=gelu	(B, 32)	2080
Dense	units=1	(B, 1)	33

Inputs and preprocessing

- Sequence: the last 30 RR, one channel, z-scored with global statistics from data/hrv_dataset.h5.
- 12 static features recomputed per window with legacy/utls2.py: rmssd, rs, ccm, ccm_n5, guzik, nn20, porta, ar_1, ar_2, ar_3, ar_4, ar_5, z-scored with data/hrv_dataset.h5 statistics.
- Target: dRR[n+1], de-normalised with the stored target mean/std (no residual over RLS).

Results

split	open loop	closed 10%	closed 30%	closed 50%	closed 80%	notes
validation (round 2)	33.72	26.07	29.43	33.65	43.88	RSA invented at 80%: 12% of subject-runs
test (final)	29.11	28.58	30.63	33.90	42.53	float only
test whole series, 80%	-	-	-	-	50.11	RSA power x1.18, invented 0.2, displaced 0.4 per subject

Correlation between the original and the filled series (validation and test subjects)

model	corr RR validation	corr RR test	corr dRR validation	corr dRR test	RMSE filled, validation (ms)	RMSE filled, test (ms)
tcn_mha_current	0.9158	0.9218	0.189	0.179	57.45	50.11
tcn_lite_ft_int8	0.9354	0.9393	0.269	0.263	50.18	43.74
rls_ar	0.9186	0.9276	0.202	0.219	56.34	49.15

Computed on the 16 validation and the 10 test subjects (results/validation/ and results/test/ folders), on the ENTIRE series, after dropping 80% of the beats (seed 7, from beat 0) and filling them in closed loop with the model (Burg-AR fill for the first 30 beats): **corr RR** = `np.corrcoef(original_serie, modified_serie)[0, 1]`; **corr dRR** = the same function on the beat-to-beat changes, `np.corrcoef(np.diff(original), np.diff(filled))[0, 1]`, which isolates the short-term dynamics. Mean over subjects; per-subject values in <folder>/crosscorr.csv. **RMSE filled**: RMSE on the model-filled beats of the same runs. Diagnostic: the selection used the round-4 protocol, not these runs.

Verdict

- Invalid for the MCU (attention, LayerNorm, GELU, global pooling, 75.5 KB, about 2M MACs).
- Beaten by tcn_lite in 16/16 validation subjects open loop and 14/16 closed loop, and by tcn_lite_ft_int8 in 10/10 test subjects open loop.
- Invents or displaces RSA peaks in the closed loop (visible in results/test/<subject>/tcn_mha_current/ spectra), the failure that started this work. The TCNs did so in at most 4% of validation subject-runs at 80% dropped and never on the test subjects.

Abbreviations and acronyms used in this document

term	meaning
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
B	batch dimension (number of windows processed together) in the layer output shapes
Burg-AR	AR model whose coefficients were fitted with Burg's method (used for the first 30 beats)
ccm	complex correlation measure: non-linearity index of the Poincare plot of consecutive RR intervals
Conv1D	one-dimensional convolution layer
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
_ft	run-name suffix: fine-tuned (training continued from the parent's weights)
GELU	Gaussian error linear unit (an activation function)
guzik	feature: Guzik asymmetry: sum of positive dRR / sum of dRR
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
KB	kilobyte (1,024 bytes)
MAC	multiply-accumulate operation (one multiplication plus one addition): the unit of compute cost on the microcontroller; MACs/beat = how many the model needs per heartbeat
MACs	multiply-accumulate operations (see MAC)
MCU	microcontroller unit
MHA	multi-head attention (a transformer layer, not supported by the MCU toolchain)
ms	milliseconds
nn20	number of successive RR differences larger than 20 ms in the window
np.corrcoef	NumPy's Pearson correlation-coefficient function
np.diff	NumPy's successive-difference function
porta	feature: Porta asymmetry: count of negative dRR / count of non-zero dRR
RLS	recursive least squares: an adaptive filter that updates its coefficients at every beat
rls_ar	baseline: adaptive per-patient AR(8) forecast by RLS
RMSE	root mean square error (ms): the main accuracy measure, lower is better
rmssd	root mean square of successive differences of the RR intervals in the window
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
rs	rescaled range: (max - min of the cumulative deviations from the mean) / standard deviation
RSA	respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing
TCN	temporal convolutional network (causal, dilated 1-D convolutions)
tcn	temporal convolutional network (run-name part)
z	z-score: (value - mean) / standard deviation, with statistics from the training subjects